Summation (Sigma) Notation

Pre-Calculus / Algebra 2 Worksheet · Grade 10–12

Name: ________________________________ Date: __________________

Learning Objectives

- Read and interpret summation (sigma) notation, identifying the lower limit, upper limit, and function
- Evaluate summation notation by substituting each integer value and adding the resulting terms
- Apply sigma notation to a variety of functions including polynomials, fractions, and expressions with multiple terms

Problems

1. Identify the lower limit, upper limit, and function in the summation expression shown.

$$\sum_{n=1}^{4} 3n$$

2. Evaluate the summation of 3n from n equals 1 to 4.

$$\sum_{n=1}^{4} 3n$$

3. Evaluate the summation of n squared from n equals 0 to 3.

$$\sum_{n=0}^{3} n^2$$

4. Evaluate the summation of the quantity x plus 3 from x equals 2 to 5.

$$\sum_{x=2}^{5} (x+3)$$

5. Evaluate the summation of 2n plus 5 from n equals 3 to 6.

$$\sum_{n=3}^{6} (2n+5)$$

---

6. Evaluate the summation of the quantity 2 divided by h from h equals 1 to 4.

$$\sum_{h=1}^{4} \frac{2}{h}$$

---

7. Evaluate the summation of the quantity 3n squared minus 1 from n equals 1 to 4.

$$\sum_{n=1}^{4} (3n^2 - 1)$$

---

8. Write the following sum using summation notation: 4 plus 8 plus 12 plus 16 plus 20.

$$4 + 8 + 12 + 16 + 20$$

---

9. Evaluate the summation of the quantity n cubed plus 2n from n equals 1 to 4.

$$\sum_{n=1}^{4} (n^3 + 2n)$$

---

10. Given the two summation expressions shown, find the value of A minus B.

$$A = \sum_{n=1}^{5} (n^2 + 1) \quad \text{and} \quad B = \sum_{n=1}^{4} (2n + 3)$$

Summation (Sigma) Notation — Answer Key

Pre-Calculus / Algebra 2 Worksheet · Grade 10–12

Answer Key

1. Answer: Lower limit: 1, Upper limit: 4, Function: 3n

- The number at the bottom of the sigma symbol ($n = 1$) is the lower limit.
- The number at the top of the sigma symbol (4) is the upper limit.
- The expression to the right of the sigma symbol ($3n$) is the function.

2. Answer: 30

- Substitute $n = 1, 2, 3, 4$ into $3n$: $3(1) + 3(2) + 3(3) + 3(4)$
- Compute each term: $3 + 6 + 9 + 12$
- Add all terms: $3 + 6 + 9 + 12 = 30$

3. Answer: 14

- Substitute $n = 0, 1, 2, 3$ into n^2: $0^2 + 1^2 + 2^2 + 3^2$
- Compute each term: $0 + 1 + 4 + 9$
- Add all terms: $0 + 1 + 4 + 9 = 14$

4. Answer: 26

- Substitute $x = 2, 3, 4, 5$ into $(x + 3)$: $(2+3) + (3+3) + (4+3) + (5+3)$
- Compute each term: $5 + 6 + 7 + 8$
- Add all terms: $5 + 6 + 7 + 8 = 26$

5. Answer: 56

- Substitute $n = 3, 4, 5, 6$ into $(2n + 5)$: $(2\cdot3+5) + (2\cdot4+5) + (2\cdot5+5) + (2\cdot6+5)$
- Compute each term: $11 + 13 + 15 + 17$
- Add all terms: $11 + 13 + 15 + 17 = 56$

6. Answer: 25/6 (approximately 4.17)

- Substitute $h = 1, 2, 3, 4$ into $2/h$: $2/1 + 2/2 + 2/3 + 2/4$
- Simplify each term: $2 + 1 + 2/3 + 1/2$
- Find a common denominator (6): $12/6 + 6/6 + 4/6 + 3/6 = 25/6 \approx 4.17$

7. Answer: 86

- Substitute $n = 1, 2, 3, 4$ into $(3n^2 – 1)$: $(3\cdot1–1) + (3\cdot4–1) + (3\cdot9–1) + (3\cdot16–1)$
- Compute each term: $2 + 11 + 26 + 47$
- Add all terms: $2 + 11 + 26 + 47 = 86$

8. Answer: Sum from n=1 to 5 of 4n

- Identify the pattern: each term is a multiple of 4 — specifically $4\cdot1, 4\cdot2, 4\cdot3, 4\cdot4, 4\cdot5$.
- The function is $4n$, the lower limit is $n = 1$, and the upper limit is $n = 5$.
- Write in sigma notation: $\sum$(n=1 to 5) 4n

9. Answer: 120

- Substitute n = 1, 2, 3, 4 into $(n^3 + 2n)$:
- n=1: 1 + 2 = 3; n=2: 8 + 4 = 12; n=3: 27 + 6 = 33; n=4: 64 + 8 = 72
- Add all terms: 3 + 12 + 33 + 72 = 120

---

10. Answer: A – B = 35

- Evaluate A = ∑(n=1 to 5)(n^2+1): (1+1)+(4+1)+(9+1)+(16+1)+(25+1) = 2+5+10+17+26 = 60
- Evaluate B = ∑(n=1 to 4)(2n+3): (2+3)+(4+3)+(6+3)+(8+3) = 5+7+9+11 = 32
- Subtract: A – B = 60 – 32 = 28
- Wait — recheck A: 2+5+10+17+26 = 60. Recheck B: 5+7+9+11 = 32. A – B = 60 – 32 = 28.
- A – B = 28
